

ANTIBACTERIAL DRUGS

(Quinolones)

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OVERVIEW

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- ▶ DNA topology in all organisms is regulated by a family of enzymes referred to as DNA topoisomerases
- ▶ Every organism requires at least type I and type II topoisomerase enzymes to manage DNA supercoiling during DNA transcription and replication
- ▶ There are two type II topoisomerases responsible for maintaining DNA in a stable and biologically active form, one of which is called DNA gyrase and the other is called topoisomerase IV
- ▶ DNA gyrase is very significant to an extent that without it, DNA replication cannot proceed
- ▶ DNA gyrase produces supercoiling by breaking the double-stranded DNA, moving a section of double-stranded DNA, and re-sealing the broken strands of DNA

Examples of quinolones

- ▶ Nalidixic acid
 - ▶ Ciprofloxacin
 - ▶ Norfloxacin
 - ▶ Ofloxacin
 - ▶ Levofloxacin
 - ▶ Moxifloxacin
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- ▶ The original quinolone nalidixic acid had limited antimicrobial activity
 - ▶ The modification of nalidixic acid by adding a fluorine atom and other moieties resulted in a group of fluoroquinolones which have an improved antibacterial activity
 - ▶ The original fluoroquinolones like ciprofloxacin are primarily active against gram negative bacteria while the newer agents like levofloxacin are active against both gram negative and gram positive organisms

Mechanism of action of quinolones

- ▶ Quinolones inhibit the two topoisomerases, the DNA gyrase and the topoisomerase IV
- ▶ Generally, DNA gyrase is the target of quinolones in gram negative bacteria while the topoisomerase IV is the primary target in gram positive organisms

Indications

- ▶ Ciprofloxacin is effective against gram negative bacteria like enteric gram negative bacilli, gonococci, chlamydia, *P.aeruginosa* and hence is used to treat infections like UTI, prostatitis and PID
- ▶ Fluoroquinolones are used to treat diarrhea infections caused *Campylobacter*, *Salmonella*, *Shigella* species including *Yersinia enterocolitica*
- ▶ Effective against traveler's diarrhea caused by enterotoxigenic strains of *E.coli*
- ▶ Fluoroquinolones are used in combination with other drugs to treat intrabdominal, bone and skin infections
- ▶ More advanced respiratory fluoroquinolones like levofloxacin, moxifloxacin and gemifloxacin have good activity against pneumococci and they are used to treat respiratory tract infections including sinusitis, bronchitis caused by *H.influenza* or *M.catarrhalis* and also effective against community acquired pneumonia
- ▶ Fluoroquinolones are active against mycobacterium and used in mycobacterium avium intracellular infections and multi drug resistant tuberculosis

Adverse effects

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- ▶ Tendonitis and tendon rupture – has increased affinity for the cartilage and tendons where they cause direct toxicity and hence should not be used in children, adolescents, pregnant women and nursing mothers
- ▶ Premature closure of the epiphyseals and thus likely to lead to stuntedness, this is another reason why these drugs are contraindicated in children
- ▶ This effect of cartilage and tendon rupture is increased in those aged 60 and above, those taking corticosteroids, those with kidney, heart or lung transplants
- ▶ Prolongs QT interval (A measure of time between the start and the end of the T wave in the heart electrical cycle)
- ▶ Alterations in blood sugar (Hypoglycemia or hyperglycemia)
- ▶ Seizures
- ▶ Phototoxicity (Dermatitis after exposure)
- ▶ Fluoroquinolones like ciprofloxacin and norfloxacin are known cytochrome P450 1A2 and hence inhibits metabolism of drugs like caffeine, theophylline etc

OTHER ANTIBACTERIAL DRUGS

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NITROFURANTOIN

- ▶ This is a synthetic nitrofurantoin derivative which is administered orally and rapidly excreted in urine
- ▶ Because of low plasma concentrations, its antibacterial activity is limited to the urinary bladder

Mechanism of action

- ▶ Nitrofurantoin is converted by nitro reductases to electrophilic intermediates which inhibits the citric cycle as well as the synthesis of DNA, RNA and protein

Indication of nitrofurantoin

- ▶ Exhibits bactericidal against gram positive and gram negative bacteria that commonly cause acute lower UTIs including *E.coli*, *Staphylococcus saprophyticus*, *Enterococcus faecalis*, *Klebsiella.pneumonia*

Adverse effects

- ▶ Nitrofurantoin is well tolerated but it can cause GIT irritation, nausea, vomiting and diarrhea
- ▶ Ingesting nitrofurantoin with food enhances its absorption and reduce the risk of GIT irritation
- ▶ Less common adverse effects are pulmonary fibrosis, hepatitis and hematologic toxicity

END